Prasanna Sattigeri

Principal Research Scientist and Manager, IBM Research · Principal Investigator, MIT-IBM Watson AI Lab

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SUMMARY

I lead post-training for agent reliability at IBM Research. The team trains Granite Guardian, an open judge model, and Granite Switch, a Granite model composed from adapter functions that check its own work. Research in generative modeling, uncertainty, causality and fairness, and agents. Principal investigator on joint projects with MIT and RPI. 10,043 citations, h-index 44 (Google Scholar, October 2026).

EXPERIENCE

IBM Research and MIT-IBM Watson AI Lab · Principal Research Scientist and Manager

2015 – present

- Lead the Granite Guardian team through five releases in 18 months, from 3.0 (October 2024) to 4.1 (April 2026): training data, post-training, evaluation and open-source release.
- Granite Guardian 4.1 8B: fifth of 20 guardrail configurations on the Artificial Analysis benchmark (F1 0.859, 59 ms), first among models that take their policy at runtime. As a reward model, 70.29 on JETTS best-of-N selection, ahead of reward models up to 70B. Six of the top ten places on GuardBench (April 2025). Third of 39 on the LLM-AggreFact leaderboard.
- Lead post-training for Granite Switch: adapter functions for policy compliance, factuality detection and correction, requirement checking, calibrated uncertainty and context attribution. Requirement checking raises IFEval balanced accuracy from 0.569 (prompted base model) to 0.855.
- Adoption: used in IBM watsonx.ai and watsonx.governance, deployed as the guardrails in Red Hat's Ask Red Hat assistant, integrated in Traefik Hub, deepset Haystack and Mozilla.ai any-guardrail. 1.15M Hugging Face downloads and 371K Ollama pulls (October 2026).
- Principal investigator on two MIT-IBM Watson AI Lab projects, with Gregory Wornell and David Sontag, and on joint work with RPI. Thesis committee member for PhD students at MIT and RPI.
- Core contributor to AI Fairness 360, AI Explainability 360 and Uncertainty Quantification 360. Generative modeling research including DIP-VAE (ICLR 2018) and semi-supervised GANs (NeurIPS 2017).

Yelp Inc. · Software Engineer

2014 – 2015

- Machine learning for search, using natural language processing and image understanding.

Arizona State University · Research Assistant

2009 – 2014

- Sparse and low-rank representation learning; lead student researcher on an NSF sensor project.

EDUCATION

2014 Ph.D., Electrical Engineering, Arizona State University

2008 B.Tech., Electronics Engineering, National Institute of Technology, India

SELECTED PUBLICATIONS

Building a Foundational Guardrail for General Agentic Systems via Synthetic Data. *ICLR 2026, co-corresponding author*

Decocted Experience Improves Test-Time Inference in LLM Agents. *COLM 2026*

Answering the Wrong Question: Reasoning Trace Inversion for Abstention in LLMs. *ACL 2026, senior author*

Granite Guardian: Comprehensive LLM Safeguarding. *NAACL 2025, senior author*

Multi-Level Explanations for Generative Language Models. *ACL 2025, oral*

Thermometer: Towards Universal Calibration for Large Language Models. *ICML 2024*

Graph-based Uncertainty Metrics for Long-form Language Model Generations. *NeurIPS 2024*

Effective Human-AI Teams via Learned Natural Language Rules and Onboarding. *NeurIPS 2023, spotlight*

Fair Infinitesimal Jackknife: Mitigating the Influence of Biased Training Data Points Without Refitting. *NeurIPS 2022, first author*

Variational Inference of Disentangled Latent Concepts from Unlabeled Observations. *ICLR 2018*

Semi-supervised Learning with GANs: Manifold Invariance with Improved Inference. *NeurIPS 2017*

AWARDS

2025	IBM Research Outstanding Accomplishment: AI governance innovations to watsonx.governance and open source
2025	IBM Research Technical Achievement Award: Granite Guardian
2024	IBM Research Outstanding Accomplishment: Granite 3.0 release
2024	IBM Research Technical Achievement Award: Causal inference for business decision making
2020–21	Four IBM Research Technical Achievement Awards: Uncertainty quantification, robust and generalizable AI, learning with less labels, efficient AI
2019	IBM Outstanding Technical Achievement Award: Contributions to trustworthy AI

PATENTS AND SERVICE

- Seven granted US patents and six applications.
- Associate Editor, Pattern Recognition. Area chair and senior program committee: NeurIPS, ICML, ICLR, AAAI.
- IEEE Senior Member.